

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Supplementary Winter Examination – 2025****Course: B TECH****Branch : MECHANICAL ENGINEERING****Semester :VI****Subject Code & Name: BTME605A_Y19 Quantitative Techniques in Project Management****Max Marks: 60****Date:****Duration: 3 Hr.****Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Operations Research is primarily concerned with:		1
	a. Theoretical physics b. Decision-making using analytical methods c. Artistic design d. History of mathematics		
2	A constraint in LPP can be represented as:		1
	a. An equality b. An equality c. Both equality and inequality d. None of the above		
3	In the North-West Corner Method, the first allocation is made in		1
	a. The center of the matrix b. The top-right corner c. The top-left corner d. The bottom-left corner		
4	In the Least Cost Method, allocations are made based on		1
	a. The highest cost b. The lowest cost c. The total distance d. The total number of shipments		
5	The Stepping Stone Method is used to		1
	a. Assign transportation routes b. Evaluate the optimality of the transportation plan c. Solve assignment problems d. Both a and b		
6	The MODI method involves the following steps		1
	a. Assigning initial allocations based on costs b. Solving a linear system of equations c. Checking the optimality of the current solution d. None of the above		

7.	Queuing theory primarily deals with			1
	a. Minimizing production costs	b. Analyzing waiting lines or queues	c. _Analyzing waiting lines or queues	d. Maximizing profit
8	The number of servers in a queuing model is represented by			1
	a. M	b. I	c. N	d. K
9	In replacement models for items that fail suddenly, if the expected cost of repair exceeds the cost of replacement, the item should be:			1
	a. _ Repaired	b. Replaced	c. Stored	d. Upgraded
10	Inventory control helps in managing:			1
	a. Raw materials	b. Work-in-progress	c. Finished goods_	d. All of the above
11	In the purchase model without shortages , if demand increases, the EOQ will:			1
	a. Increase	b. Decrease	c. Stay the same	d. Not be affected
12	The EOQ with price breaks model accounts for:			1
	a.A constant price per unit	b. A discount based on order quantity	c. A varying lead time for different quantities	d. Both b and c
Q. 2	Solve the following.			12
A)	Explain the process of formulating a Linear Programming Problem (LPP) from a given real-world scenario. Provide an example of such a formulation.			6
B)	Describe the Simplex Method for solving Linear Programming Problems. What are the key steps in this method, and how does it help in obtaining the optimal solution for problems with more than two decision variables?			6
Q.3	Solve the following.			12
A)	Explain the Transportation Problem in Operations Research. Describe the different methods used to find an initial feasible solution to a transportation problem, including the North West Corner Method, Least Cost Method, and Vogel's Approximation Method (VAM).			6
B)	Describe the Traveling Salesman Problem (TSP). Discuss the brute force method and nearest neighbor method as approaches to solve the TSP.			6
Q. 4	Solve Any Two of the following.			12
A)	Discuss the M/M/I (FCFS) model of queuing, which is also known as the Birth and Death Model . Explain the key assumptions, the underlying mathematical concepts, and how it can be used to analyze a system with an infinite number of customers and servers.			6

B)	Explain the concept of Inventory Management . Discuss its significance in ensuring the smooth flow of materials in production, and how it helps in minimizing costs while maximizing customer satisfaction.		6
C)	Explain the difference between a project and other manufacturing systems . How does project-based work differ from traditional manufacturing operations in terms of structure, objectives, and management?		6
Q.5	Solve Any Two of the following.		12
A)	Explain the necessity of different planning techniques for project management . Why are various methods like Gantt charts, PERT, and CPM required for effective project execution?		6
B)	Explain the Critical Path Method (CPM) and its role in project management. How is the critical path determined, and why is it important for controlling project timelines?		6
C)	Describe the Program Evaluation and Review Technique (PERT) . How does PERT differ from CPM, and what is its use in managing projects with uncertain task durations?		6
Q. 6	Solve Any Two of the following.		12
A)	Explain the importance of time and cost estimates in project management. How do accurate time and cost estimates contribute to the success of a project?		6
B)	Explain the concept of crashing the project duration . How does crashing affect the cost of the project , and what are the key factors that need to be considered before deciding to crash a project?		6
C)	Discuss the impact of resource constraints on project duration and cost . How do project managers balance resource availability with the need to meet deadlines and budget constraints?		6
	*** End ***		